PROJECT DESIGN PHASE

Solution Architecture

Date	03 October 2026
Team ID	SWTID-2026-6390
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Solution Architecture

Script-Controlled ACL – Restrict Record Access Based on Field Value

Architecture Overview

The whole solution is built inside the ServiceNow platform. It uses a custom table called Institution Details (u_institution_details), four custom roles (bb1 to bb4), record-level Access Control Lists, a Branch = EEE condition and an advanced script for the READ ACL. The design keeps two things apart: first the system finds out which roles the user has, and then each operation (read, create, write, delete) is checked by its own ACL. READ is the only ACL that also looks at the Branch value of the record. CREATE, WRITE and DELETE depend on bb2, bb3 and bb4. The administrator always has unrestricted access.

Architecture Flow

User Login	The user signs in to ServiceNow.
↓	
Role Identification	ServiceNow finds the roles given to the user (admin, bb1, bb2, bb3, bb4).
↓	
Institution Details Request	The user opens the list or tries to read, create, edit or delete a record in u_institution_details.
↓	
ACL Evaluation	ServiceNow picks the ACL that matches the operation.
↓	
Branch / Role Validation	For READ, the Branch = EEE condition and the advanced script are checked. For the other operations, the required role (bb2, bb3 or bb4) is checked.
↓	
Allow or Deny Access	If the ACL conditions are satisfied the action is allowed. Otherwise it is denied.

Roles and Their Permissions

Role	Operation	Meaning
bb1	READ	Can read Institution Details records that satisfy the READ ACL (Branch = EEE).
bb2	CREATE	Can create new records.
bb3	WRITE	Can edit existing records.
bb4	DELETE	Can delete records.
admin	All	Unrestricted access to all records and operations.

ACL Configuration

Operation	ACL	Role	Condition / Logic	Expected Result
READ	u_institution_details (read)	bb1	Branch = EEE, plus advanced script. Administrator is allowed.	bb1 users see EEE records. Administrator sees all records.

Operation	ACL	Role	Condition / Logic	Expected Result
CREATE	u_institution_details (create)	bb2	No extra data condition.	Users with bb2 can create records.
WRITE	u_institution_details (write)	bb3	No extra data condition.	Users with bb3 can edit records they have access to.
DELETE	u_institution_details (delete)	bb4	No extra data condition.	Users with bb4 can delete records they have access to.

READ ACL Script Logic

```
(function () {
    if (gs.hasRole('admin')) {
        return true;
    }
    if (gs.hasRole('bb1')) {
        return true;
    }
    return false;
})();
```

- Administrator → allow.
- bb1 → allow.
- Anyone else → deny.
- The Branch = EEE data condition of the READ ACL limits which records apply, so bb1 users get EEE records only.

End-to-End Flow

1. The user logs into ServiceNow.
2. ServiceNow identifies the roles assigned to the user.
3. The user opens Institution Details or performs an operation on a record.
4. ServiceNow evaluates the ACL for that operation (READ, CREATE, WRITE or DELETE).
5. For READ, the Branch = EEE condition and the advanced script are also checked.
6. The action is allowed if the ACL is satisfied; otherwise access is denied.
7. Impersonation is used to confirm this behaviour for an EEE user, users with other roles, and the administrator.

Key Components

Component	Purpose
ServiceNow platform	Hosts the table, users, roles, ACLs, lists and forms.
u_institution_details	Custom table (Institution Details) with Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description.
Branch field	Choice field with ECE, EEE and CSE; used by the READ condition.
Roles bb1–bb4	bb1 = READ, bb2 = CREATE, bb3 = WRITE, bb4 = DELETE.
ACL engine	Checks the matching ACL on the server before an operation is allowed.
READ script	Advanced script using gs.hasRole() that allows admin and bb1 and denies others.
Impersonation testing	Used to check EEE-only read access and the CREATE, WRITE and DELETE permissions.